THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The electronic configuration of oxygen is: 1

(a) 2, 8 **(b)** 2, 5 **(c)** 2, 6 **(d)** 2, 4

2. The isotope used for cancer treatment is: 1

(a) chlorine-35 **(b)** cobalt-60 **(c)** carbon-12 **(d)** oxygen-16

3. The atomic number is the number of: 1

(a) neutrons **(b)** electrons only **(c)** nucleons **(d)** protons

4. The electronic configuration of neon is: 1

(a) 2, 7 **(b)** 2, 6 **(c)** 8, 2 **(d)** 2, 8

5. The isotope used for goitre is: 1

(a) cobalt-60 **(b)** carbon-12 **(c)** iodine-131 **(d)** uranium-235

6. The mass number is the number of: 1

(a) neutrons only **(b)** protons + neutrons

(c) protons only **(d)** electrons

7. The valency of sodium is: 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 3

8. The fuel in nuclear reactors is: 1

(a) carbon-14 **(b)** sodium-23 **(c)** uranium-235 **(d)** oxygen-16

9. Rutherford discovered the: 1

(a) proton **(b)** electron **(c)** nucleus **(d)** neutron

10. The valency of oxygen is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What are isotopes? 2

12. Name one use of an isotope. 2

13. What is atomic number? 2

14. What are isobars? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the number of p, e, n in $^{12}_{6}\mathrm{C}$. **3**

16. Give the valencies of H, Li, F. **3**

17. State three postulates of Bohr's model. **3**

OR

Find the number of p, e, n in $^{16}_{8}\mathrm{O}$. **3**

18. Give the valencies of Mg, Ca, O. **3**

OR

An atom has 17 protons and 18 neutrons. Find the mass number and the electronic configuration. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe Bohr's model of the atom. **5**

OR

Explain valency with examples. **5**

20. Explain the gold foil experiment. **5**

OR

Complete the table for $^{12}_{6}\mathrm{C}$, $^{14}_{7}\mathrm{N}$, $^{16}_{8}\mathrm{O}$, $^{19}_{9}\mathrm{F}$. **5**

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND